// Day 21 of 30

VISUAL FLOWCHART

Spring Boot Request Lifecycle

→ Swipe through all 6 slides

Shubham Bhati
linkedin.com/in/bhatishubham

Step 1: Tomcat Receives Request

Embedded server catches HTTP

```
HTTP Request arrives at port 8080

Embedded Tomcat:
  → Accepts TCP connection
  → Parses HTTP (method, headers, body)
  → Creates HttpServletRequest object
  → Hands off to Spring's DispatcherServlet
```

💡 Tomcat is embedded in Spring Boot — no separate install

Step 2: Filter Chain

Security, CORS, logging — before your code

```
Request passes through filters:

1. CorsFilter → CORS headers
2. SecurityFilter → JWT validation
3. LoggingFilter → request logging
4. Custom filters you add

If auth fails → 401 returned here
Request never reaches your controller
```

💡 Filters run BEFORE your controller code

Step 3: DispatcherServlet

The traffic cop of Spring MVC

```
DispatcherServlet:
  → Receives request from filters
  → Asks HandlerMapping:
    'Which controller handles GET /users/1?'
  → HandlerMapping returns:
    UserController.getUser()
  → Dispatches request to controller
```

💡 DispatcherServlet = central router for all requests

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linkedin.com/in/bhatishubham

Step 4: Controller → Service → Repo

Your code runs here

```
@GetMapping("/users/{id}")
public UserDTO getUser(@PathVariable Long id) {
  return userService.findById(id);
}

→ Service layer: business logic
→ Repository layer: database query
→ Hibernate: generates SQL
→ MySQL: returns data
→ Entity → DTO → returned
```

💡 Controller → Service → Repository. Always.

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linkedin.com/in/bhatishubham

Step 5: Response Serialization

Java object → JSON

```
UserDTO object returned from controller

Jackson (default serializer):
  → Converts Java object to JSON
  → Respects @JsonIgnore, @JsonProperty
  → Sets Content-Type: application/json

{
  "id": 1,
  "name": "Shubham",
  "email": "shubham@test.com"
}
```

💡 Jackson handles all serialization automatically

Step 6: Response Sent

Back through the chain

```
Response travels BACK through:
  → Exception handlers (if error)
  → Response filters
  → Tomcat sends HTTP response

HTTP/1.1 200 OK
Content-Type: application/json
{ ... }

Total time: ~5-50ms typically
```

💡 Understanding this lifecycle = understanding Spring Boot

Shubham Bhati
linkedin.com/in/bhatishubham

```
done = true
```

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Shubham Bhati
linkedin.com/in/bhatishubham